

BASIC COMMANDS

TO CHECK THE PRESENT WORKING LOCATION

```
# pwd
```

TO CHECK THE CONTENTS OF DIRECTORY

```
# ls
```

TO CHECK THE PROPERTIES OF FILES & DIRECTORIES

```
# ll
```

TO CHECK THE HIDDEN FILES & DIRECTORIES

```
# ls -a
```

TO CHECK THE INODE NUMBER OF FILES & DIRECTORIES

```
# ls -i
```

TO CREATE A FILE WITH DATA

```
# cat > (filename)
```

```
# cat > linux
```

```
    This is linux
```

```
    Press ctrl+D to save & quit the file
```

```
# cat linux (to read the file)
```

TO APPEND THE DATA WITHOUT OVERWRITING

```
# cat >> linux
```

```
    Added data by root
```

```
    Press ctrl+D to save & quit the file
```

```
# cat linux (read the file)
```

TO CREATE MULTIPLE FILES WITHOUT DATA

```
# touch unix solaris redhat
```

```
# ls (to check)
```

TO CREATE A DIRECTORY

```
# mkdir <dir_name>
```

```
# mkdir zoom
```

```
# ls (to check)
```

```
# cd zoom (to enter into dir)
```

```
# cd (to come out)
```

TO CREATE A MULTIPLE DIRECTORIES

```
# mkdir zoom1 zoom2 zoom3 zoom4
```

```
# ls (to check)
```

TO CREATE A PARENT DIRECTORY

```
# mkdir -p world/asia/india/ap/hyd/dsnr/zoom
# ls -R world (to check)
```

NAVIGATION OF DIRECTORIES

```
# cd world/asia/india/ap/hyd/dsnr/zoom
# pwd (to check)
# cd .. (to come 1 step back)
# cd ../.. (to come 2 step back)
# cd ../../.. (to come 3 step back)
# cd - (to go at last working directory)
# cd (to go at user home directory)
# pwd (to check)
```

TO CHECK AND CHANGE THE DATE & TIME

```
# date (to check)
# date -s "wed Mar 19 12:00:00 IST 2014"
```

TO CHECK THE CALENDER

```
# cal (for current month & year)
# cal 2014 (for year)
# cal 5 2014 (for particular month & year)
```

TO OPEN THE CALCULATOR

```
# bc
1+2+3
7
Type 'q' to come out
```

TO GET THE HELP RELATED TO ANY COMMAND

```
# man <command name>
# man cat (for example)
Type 'q' to come out
```

TO SHUTDOWN THE SYSTEM

```
# init 0
```

TO RESTART THE SYSTEM

```
# init 6
```

TO SEARCH FILES OR DIRECTORIES

```
# find / -iname <filename or dirname>
```

TO ACCESS ANY LOCATION IN GUI

```
# nautilus <Path>
# nautilus /etc (for example)
```

TO COPY A EXISTING FILE WITH NEW NAME

```
# cp <source_file> <new_name>
# cp linux unix
# ls (to check)
# cat unix (to read)
```

TO COPY FILE INTO DIFFERENT LOCATION

```
# cp (filename) (destination)
# cp linux /opt
# ls /opt (to check)
```

TO COPY MULTIPLE FILES

```
# cp unix redhat solaris /opt
# ls /opt (to check)
```

TO COPY DATA FROM SOURCE FILE TO DESTINATION WITHOUT OVERWRITING

```
# cat (source_file) >> (destination_file)
# cat linux >> redhat
# cat redhat (to check)
```

TO COPY A DIRECTORY

```
# cp -rv (source_dir) (destination)
# cp -rv zoom /opt
# ls /opt (to check)
```

TO COPY MULTIPLE DIRECTORIES

```
# cp -rv zoom1 zoom2 zoom3 /opt
# ls /opt (to check)
```

TO COPY COMPLETE DATA FROM PRESENT WORKING LOCATION

```
# cp -rv * /media
# ls /media (to check)
```

TO CUT & PASTE FILE OR DIRECTORIES

```
# mv (source_dir or file) (destination)
# mv zoom /mnt
# ls (to check)
# ls /mnt (to check)
```

TO RENAME A FILE OR DIRECTORY

```
# mv (existing_name) (new_name)
# mv zoom1 zoomgroup
# ls (to check)
```

TO REMOVE A FILE OR DIRECTORY

```
# rm (filename)
# rm -rf (directory_name)
```

TO OPEN THE TASK MANAGER

```
# ps -aux
```

TO KILL THE PROCESS OF RUNNING APPLICATION

```
# kill -9 (process_id)
# kill -9 2354 (for example)
```

TO CHECK THE CONFIGURATION OF CPU

```
# cat /proc/cpuinfo
```

TO CHECK THE RUNNING PROCESS OF CPU

```
# top
```

TO CHECK THE OS VERSION

```
# cat /etc/redhat-release
```

TO CHECK THE LAST LOGIN TIME

```
# last
```

(1) USER ADMINISTRATION PRACTICAL STEPS

'#### TO CREATE A USER ####'

```
# useradd <username>
# useradd tom
```

'#### TO CHECK THE USER DETAILS ####'

```
# cat /etc/passwd      (for complete details)
# tail /etc/passwd     (for last 10 users details)
# tail -3 /etc/passwd  (for last 3 users details)
# head /etc/passwd     (for top 10 users details)
# head -3 /etc/passwd  (for top 3 users details)
# grep tom /etc/passwd (for particular user details)
```

'#### ASSIGNING A PASSWORD TO USER ####'

```
# passwd <username>
# passwd tom
    Password: 1
Re-Password: 1
```

'#### TO CHECK THE USER PASSWORD DETAILS ####'

```
# cat /etc/shadow
# tail /etc/shadow
```

'#### TO CHECK THE PASSWORD ENCRYPTION TOOL ####'

```
# passwd -S <username>
# passwd -S tom
```

'#### TO CHANGE THE PASSWORD ENCRYPTION TOOL ####'

```
# authconfig-tui
    By pressing spacebar key uncheck the option USE MD5
    Click on NEXT
# useradd jack (create a user)
# passwd jack  (assign the password)
# passwd -S jack (check the password tool)
```

'#### TO LOGIN WITH A USER ####'

```
# su - tom (In root Console)
# exit (to logout)
```

```
# gdmflexiserver (to open extra graphical console)
    Select the user and enter the password
    Create Some Data
    Go to root console by pressing ctrl+alt+F1
    And check the user data
```

```
# ls /home/tom/Desktop
# who (to get the details of active users)
```

```
'#### TO LOCK A USER ####'
```

```
# usermod -L <username>
# usermod -L tom
# grep tom /etc/shadow (to check)
  '!' sign indicates that a user is lock
# gdmflexiserver
  Try to login with user tom
```

```
'#### TO UNLOCK A USER ####'
```

```
# usermod -U <username>
# usermod -U tom
# grep tom /etc/shadow (to check)
# gdmflexiserver
  Try to login with user tom
```

```
'#### TO CHECK THE MAIL ACCOUNT OF USERS ####'
```

```
# cd /var/spool/mail
# ls (to check)
```

```
'#### TO CHECK THE GROUP OF THE USER ####'
```

```
# tail /etc/group
```

```
'#### TO CHECK THE AVAILABLE SHELLS ####'
```

```
# cat /etc/shells
```

(2) USER MODIFICATION PRACTICAL STEPS

'### TO ASSIGN THE COMMENT ###'

```
# useradd tom (create one user)
# tail /etc/passwd (to check)
# usermod -c <comment> <username>
# usermod -c "Sales Dept" tom
# grep tom /etc/passwd (to check)
```

'### TO CHANGE THE UID ###'

```
# useradd jack (create one more user)
# tail -2 /etc/passwd (to check)
To Provide User:jack Privileges to User:tom
# usermod -u <ID> -o <username>
# usermod -u 501 -o tom
# tail -2 /etc/passwd (to check)
Login with both the user's to check the result
```

'### TO CHANGE THE GID ###'

```
# groupmod -g <ID> <username>
# groupmod -g 600 tom
# grep tom /etc/passwd (to check)
Changing the GID means,taking the Group Privileges
```

'### TO CHANGE THE SHELL ###'

```
# cat /etc/shells (to check available shells)
# usermod -s <shell> <username>
# usermod -s /sbin/nologin tom (to block account)
# grep tom /etc/passwd (to check)
# su - tom (try to access tom account)
```

'### TO LOCK THE USER ACCOUNT ###'

```
# usermod -L <username>
# usermod -L jack (to lock)
# grep jack /etc/shadow (to check)
'!'sign indicates that the user account is lock
# gdmflexiserver (open one gui console)
Select the User:jack & try to login
# usermod -U tom (to unlock)
# grep jack /etc/shadow (to check)
# gdmflexiserver (open one gui console)
Select the User:jack & try to login
```

'### TO CHANGE THE USER HOME DIRECTORY ###'

```
# vi /etc/default/useradd (open this file)
  Do the Modification in the HOME line,as below
  HOME=/media
  :wq(save & quit the file)
# useradd jerry (create one user)
# tail /etc/passwd (to check)
```

'### TO CHANGE THE USERNAME ###'

```
# usermod -l <newname> <oldname>
# usermod -l tom123 tom
# tail /etc/passwd (to check)
```

'### TO REMOVE A USER ###'

```
# userdel jack (to remove a user without its home directory &
mail account)
# tail /etc/passwd (to check)
# ls /home (to check home dir)
# ls /var/spool/mail (to check mail accout)
# userdel -r tom123 (to remove a user with its home directory &
mail account)
# tail /etc/passwd (to check)
# ls /home (to check)
# ls /var/spool/mail (to check)
```


(3) GROUP ADMINISTRATION PRACTICAL STEPS

Create 5 Users

```
'#### TO CREATE A GROUP ####'
```

```
# groupadd <group_name>
# groupadd zoom
# tail /etc/group (to check)
```

```
'### TO ADD A SINGLE USER INTO GROUP ###'
```

```
# gpasswd -a <username> <groupname>
# gpasswd -a a1 zoom
# grep zoom /etc/group (to check)
```

```
'### TO ADD MULTIPLE USERS INTO GROUP ###'
```

```
# gpasswd -M a2,a3,a4 zoom
# grep zoom /etc/group (to check)
```

Note:-Adding Multiple Users Into the Group may get overwrite on the existing users. So Without Overwriting to add a user then use the above option(gpasswd -a)

```
'### TO REMOVE A USER FROM GROUP ###'
```

```
# gpasswd -d <username> <groupame>
# gpasswd -d a4 zoom
# grep zoom /etc/group (to check)
```

```
'### TO MAKE A USER AS GROUP ADMIN ###'
```

```
# gpasswd -A <username> <groupname>
# gpasswd -A a2 zoom
# grep zoom /etc/gshadow (to check)
# su - a2 (to check, login as a user)
# gpasswd -d a3 zoom (try to remove)
# gpasswd -a a5 zoom (try to add)
# grep zoom /etc/group (to check)
# exit (logout)
# gpasswd -A "" zoom (to remove the previlages)
# grep zoom /etc/gshadow (to check)
```

```
'### TO RENAME A GROUP ###'
```

```
# groupmod -n <newname> <oldname>
# groupmod -n zoomgroup zoom
# tail /etc/group (to check)
```

```
'### TO REMOVE A GROUP ###'
```

```
# groupdel zoomgroup
# tail /etc/group (to check)
```

(4) PERMISSIONS PRACTICAL STEPS

BASIC PERMISSION

'### APPLYING A BASIC PERMISSION ON DIRECTORY ###'

```
# useradd tom (create one user)
# su - tom (login as a user)
# cd /root (try to access root dir)
# exit (logout)
# ls -ld /root (check the permission)
```

(1) To Apply Execute Permission on Directory

```
# chmod 751 /root
# ls -ld /root (to check)
# su - tom
# cd /root (try to access)
# ls (to read - ERROR)
# exit (logout)
```

(2) To Apply Read & Execute Permission on Dir

```
# chmod 755 /root
      OR
# chmod u=rwx,g=rx,o=rx /root
# ls -ld /root (to check)
# su - tom
# cd /root
# ls (to read)
# touch tom (to write - ERROR)
# exit (logout)
```

(3) To Apply Read, Write & Execute Permission on Dir

```
# chmod 777 /root
      OR
# chmod u=rwx,g=rwx,o=rwx /root
      OR
# chmod a=rwx /root
# ls -ld /root (to check)
# su - tom
# cd /root
# ls (to read)
# touch tom (to write)
# exit (logout)
```

'#### BASIC PERMISSION ON FILE ####'

```
# cat > linux (create one file in /root dir)
# su - tom
# cd /root
# cat linux (to read)
# cat >> linux (to write - ERROR)
# exit
# ll linux (check the permission)
```

To Apply Read & Write Permission file

```
# chmod 666 linux
# ll linux (to check)
# su - tom
# cd /root
# cat linux (to read)
# cat >> linux (to write)
# exit
```

'#### TO CHANGE THE OWNER OF THE FILE ###'

Create 3 more users (for ex:a1,a2,a3)

```
# cat > unix (create one more file in /root)
# ll unix (check the permission)
For the User:a1-rw permission on unix file
# chown <username> <filename>
# chown a1 unix
# ll linux (to check)
# su - a1 (to check-login as a user)
# cd /root
# cat unix (to read)
# cat >> unix (to write)
# exit (logout)
```

'#### TO CHANGE THE GROUP OWNER OF THE FILE ####'

For the User:a2,a3-zero permission on unix file

```
# groupadd zoom (create one group)
# gpasswd -M a2,a3 zoom (add two users in group)
# chgrp <groupname> <filename>
# chgrp zoom unix (to change group owner)
# chmod 604 unix (provide zero permission in group)
# ll unix (to check)
```

To Check the Result Login As a User:a2,a3 and access the unix file in /root directory

ADVANCE PERMISSION

TO APPLY ADVANCE PERMISSION ON DIRECTORY

```
# chmod 1777 /root
'1'for applying advance (sticky bit) permission only on
directories to avoid data deleting
# ls -ld /root
't'sign is the indication of Advance Permission
# su - tom (login as a normal user)
# cd /root (access root directory)
# rm -rf * (try to remove the data)
# exit (logout)
```

ACL PERMISSION

Create 6 Users (for ex:a1 to a6)

```
# cat > linux (create one file in /root)
# ll linux (to check the permission)
```

'For the User:a1 - Read & Write Permission'

```
# setfacl -m u:a1:rw linux (to assign acl)
# ll linux (to check)
'+'sign is the indication of acl permission
# su - a1 (to check-login as a user)
# cd /root
# cat linux (to read)
# cat >> linux (to write)
# exit (logout)
```

'For the User:a2 - Read & Execute Permission'

```
# setfacl -m u:a2:rx linux
# getfacl linux (to check acl permission in detail)
Login as a User:a2 & Check the Result
```

'For the User:a3 - Read,Write & Execute Permission'

```
# setfacl -m u:a3:rwX linux
# getfacl linux (to check)
Login as a User:a3 & Check the Result
```

'For the User:a4 - Write Permission'

```
# setfacl -m u:a4:w linux
# getfacl linux (to check)
Login as a User:a4 & Check the Result
```

'For the User:a5,a6 - Zero Permission'

```
# groupadd zoom (create one group)
# gpasswd -M a5,a6 zoom
# setfacl -m g:zoom:0 linux (to assign on group)
# getfacl linux (to check)
Login as a user:a5,a6 & check the result
```

'To Remove the ACL Permission from single user'

```
# setfacl -x u:a4 linux (to remove)
# getfacl linux (to check)
```

'To Remove the ACL Permission Completely from file'

```
# setfacl --remove-all linux
# getfacl linux (to check)
# ll linux (to check)
```

(5) PARTITIONS PRACTICAL STEPS

'Step 1:-Check the Existing Partitions'

```
# fdisk -l (to check)
```

'Step 2:-To create a New Partition'

```
# fdisk /dev/sda
Press 'm' for help
Press 'p' to check the existing partitions
Press 'n' to create a new partition
Enter the Size: +1G (for example)
Press 'p' to check
Press 'w' to save & quit
```

'Step 3:-Activate the New Changes to Kernel'

```
# partx -a /dev/sda (run this 2 times)
```

'Step 4:-Format the Partition'

```
# mkfs.ext4 <partition_no>
# mkfs.ext4 /dev/sda8 (for example)
```

'Step 5:-To Use the Partition-Create Mount Point'

```
# mount <partition_no> <mount_point>
# mount /dev/sda8 /mnt
# mount (to check)
# cd /mnt (to access the partition)
# ls (to check)
'lost+found' dir is the indication of partition
# mkdir zoom{1..100} (create some data)
# ls (to check)
```

'#### TO CHECK THE FILESYSTEM OF PARTITIONS ####'

```
# blkid
```

'#### TO CHECK THE FREE SPACE OF THE PARTITION ###'

```
# df -h
```

'#### TO CHECK THE USED SPACE OF THE PARTITION ####'

```
# du -sh
```

'#### TO CHECK THE FREE SPACE OF THE HARDISK ####'

```
# parted
Type: print free
Type: quit (to come out)
```

```
'#### TO REMOVE THE MOUNT POINT(LINK) ####'
```

```
# umount <partition_no>  
# umount /dev/sda8 (for example)  
# mount (to check)
```

```
'#### TO DO THE PERMANENT MOUNTING ####'
```

```
# vim /etc/fstab (open this file)  
Come at last & do the changes as given below
```

```
/dev/sda11      /mnt            ext4           defaults 0 0  
'partition_no' 'mount_pt' 'filesystem'
```

```
:wq(save & quit the file)  
# init 6 (reboot the pc)  
# mount (to check the permanent mount point)
```

(6) LVM PRACTICAL STEPS

TO IMPLEMENT LVM ON PARTITIONS

'Step 1:-Create 3 Raw Partitions'

```
# fdisk -l (to check)
# fdisk /dev/sda (to create-see partition notes)
# partx -a /dev/sda (to update kernel)
```

'Step 2:-Create Physical Volume'

```
# pvcreate <partition_no's>
# pvcreate /dev/sda{12..14} (for example)
# pvdisplay (to check)
```

'Step 3:-Create Volume Group'

```
# vgcreate <vg_name> <partition_no's>
# vgcreate zoom /dev/sda{12..14}
# vgdisplay (to check)
```

'Step 4:-Create Logical Volume'

```
# lvcreate -L <size> <vg_name> -n <lv_name>
# lvcreate -L +500M zoom -n linux
# lvdisplay (to check)
```

'Step 5:-Format LV & Mount to Use'

```
# mkfs.ext4 /dev/zoom/linux
# mkdir /lvm (create new dir for mount point)
# mount /dev/zoom/linux /lvm (to mount)
# mount (to check)
# cd /lvm (access mount point to use LV)
# mkdir zoom{1..100} (create some data)
# ls (to check)
# df -h (to check the free space of partition)
```

'Step 6:-To Resize the Partition'

```
# cp -rv /usr/* /lvm (to make partition size full)
# df -h (check the size)
# lvresize -L +500M /dev/zoom/linux (to resize)
# resize2fs /dev/zoom/linux (to resize filesystem)
# df -h (check the extended size)
```


'Step 7:-To Extend the Volume Group Size'

```
# vgdisplay (to check vg size)
# fdisk /dev/sda (create one partition)
# partx -a /dev/sda (to update)
# pvcreate /dev/sda15 (add in pv)
# vgextend zoom /dev/sda15 (to extend vg size)
# vgdisplay (to check extended vg size)
```

'Step 8:-To Remove LVM Concept from Partition'

```
# umount /lvm (un mount the lv)
# lvremove /dev/zoom/linux (1st remove lv)
# lvdisplay (to check)
# vgremove /dev/zoom (2nd remove vg)
# vgdisplay (to check)
# pvremove /dev/sda{12..15} (3rd remove pv's)
# pvdisplay (to check)
```

(7) SWAP PARTITION PRACTICAL STEPS

TO CREATE A SWAP PARTITION

'Step 1:-Check the RAM size'

```
# free -m (to check the RAM size in MB)
```

'Step 2:-Create One Partition'

```
# fdisk /dev/sda (to create)
  Press 'n' to create new partition
  Enter the size: +4G (for example 4GB)
  Press 't' to change the ID of Partition
  Enter Partition No:16 (for example)
  Press 'L' to get the list of available ID's
  Enter 82 for swap
  Press 'p' to check
  Press 'w' to save & quit
```

'Step 3:-Update & Format the Partition'

```
# partx -a /dev/sda (to update)
# mkswap /dev/sda16 (to create swap filesystem)
# blkid /dev/sda16 (to check)
```

'Step 4:-Enable the Swap Service'

```
# swapon /dev/sda16 (to enable)
# free -m (to check RAM size)
# swapon -s (to check the status of swap partition)
```

'Step 5:-To Disable the Swap Service'

```
# swapoff /dev/sda16 (to disable)
# free -m (to check the RAM size)
# swapon -s (to check the status)
```

(8) RAID PRACTICAL STEPS

TO IMPLEMENT RAID 5 ON PARTITIONS

'Step 1:-Create 3 Raw Partitions for Raid 5'

```
# fdisk /dev/sda (to create)
# partx -a /dev/sda (to update kernel)
```

'Step 2:-Create a Metadisk'

```
# mdadm -C <metadisk_name> -n3 <partition_no's> -l5
# mdadm -C /dev/md0 -n3 /dev/sda{15..17} -l5
# mdadm -D /dev/md0 (to check)
```

'Step 3:-Format & Mount the Metadisk to Use'

```
# mkfs.ext4 /dev/md0 (to format)
# mkdir /raid (create new mount point)
# mount /dev/md0 /raid (to mount)
# mount (to check)
# cd /raid (to access the metadisk)
# mkdir zoom{1..100} (create some data)
# ls (to check)
# cd (come out)
```

'Step 4:-To check Raid-5 is Working or Not-Then in this case make one partition faulty & remove'

```
# mdadm -D /dev/md0 (to check the status)
# mdadm -f /dev/md0 /dev/sda15 (to make faulty)
# mdadm -D /dev/md0 (to check)
# mdadm -r /dev/md0 /dev/sda15 (to remove)
# mdadm -D /dev/md0 (to check)
# ls /raid (to check the data in metadisk)
```

'Step 5:-To work with Raid-5,add one Partition'

```
# fdisk /dev/sda (to create)
# partx -a /dev/sda (to update kernel)
# mkfs.ext4 /dev/sda18 (for example-to format)
# mdadm -a /dev/md0 /dev/sda18 (to add in metadisk)
# mdadm -D /dev/md0 (to check)
# ls /raid (check the data)
```

'Step 6:-To Stop the Raid Level 5'

```
# umount /raid (unmount the metadisk)
# mdadm -S /dev/md0 (to stop)
# mdadm -D /dev/md0 (to check)
# ls /raid (to check data in mount point)
```

'Step 7:-To Start the Raid Level 5'

```
# mdadm -A /dev/md0 /dev/sda{16..18} (to start)
# mdadm -D /dev/md0 (to check)
# mount /dev/md0 /raid (to access metadisk)
# ls /raid (to check the data in metadisk)
```

(9)DISK QUOTAS PRACTICAL STEPS

TO IMPEMMENT THE DISK QUOTAS ON PARTITION

'Step 1:-Create One Partition & Format'

```
# fdisk /dev/sda (to create)
# partx -a /dev/sda (to update)
# mkfs.ext4 /dev/sda8 (for example-to format)
# mkdir /quota (create a new mount point)
```

'Step 2:-Apply Quotas Permission on the Partition'

```
# mount -o usrquota,grpquota /dev/sda8 /quota
# mount (to check)
```

'Step 3:-Create Quota Database in the Partition'

```
# quotacheck -cugv /quota
```

'Step 4:-Enable the Quota Service'

```
# quotaon /quota (to enable)
# quotaon -p /quota (to check)
```

'Step 5:-Apply Full Permission on Mount Point'

```
# chmod 777 /quota
```

'Step 6:-Define the Quota Limit Based on Inode or Block'

Create 2 Users

```
# useradd tom
# useradd jack
```

'### Defining Quotas Based on User Level ###'

(A)Implemmenting quotas based on inode for User:tom

```
# edquota -u tom
  Filesystem    block soft hard   inode soft hard
  /dev/sda8         0     0   0         0     5   10
:wq(save & quit the file)
# su - tom (login as a user to check the result)
# cd /quota (access mount point)
# touch tom{1..10} (create 10 files)
# ls (to check)
# touch tom11 (try to create 11th file)
# exit (logout)
```

(B) Implementing quotas based on block for User:jack

```
# blockdev --getbsz /dev/sda8 (to check block size)
# edquota -u jack
  Filesystem    block soft hard    inode soft hard
  /dev/sda8      0   12   20      0     0     0
:wq(save & quit the file)
# su - jack (login as a user to check the result)
# cd /quota (access mount point)
# mkdir jack{1..5} (create 5 dir's)
# ls (to check)
# mkdir jack6 (try to create 6th dir)
# exit (logout)
NOTE:-If the block size of the partition is 4096 bytes
      then the size of each empty dir will be 4kb
      (1 block=1kb)
```

'Defining the Quotas Based on Group Level'

Create 3 Users (for example:a1 to a3)

```
# groupadd zoom (create one group)
# usermod -g zoom a1 (to add user as a primary)
# usermod -g zoom a2
# usermod -g zoom a3
```

(A) Implementing quotas based on inode for group:zoom

```
# edquota -g zoom
  Filesystem    block soft hard    inode soft hard
  /dev/sda8      0     0     0      0    10    15
:wq(save & quit the file)
# su - a1
# cd /quota
# touch file{1..5} (create 5 files as user:a1)
# exit (logout)
Then Create 5 files each as a user:a2 & a3
# touch file16 (try to create 16th file as user:a3)
# exit (logout)
```

(10) BACKUP PRACTICAL STEPS

BACKUP & RECOVERY

'TO TAKE A BACKUP FROM TAR PROGRAM - FOR DIRECTORIES'

```
# mkdir zoom (create one dir /root)
# touch zoom/liux{1..10} (add some data)
# ls zoom (to check)
```

TO TAKE BACKUP:-

```
# tar -cvf <destination/backupfile> <source>
# tar -cvf /opt/zoom.tar zoom
```

```
# cd /opt
# ls (to check)
TO READ BACKUP FILE:-
# tar -tvf zoom.tar
# cd (go to original & remove data)
# rm -rf zoom
# ls (to check)
```

```
TO RESTORE:-
# tar -xvf /opt/zoom.tar
# ls (to check)
```

'TO TAKE A BACKUP FROM CPIO PROGRAM - FOR FILES'

```
# touch linux unix redhat (create some files in /root)
# ls (to check)
```

```
TO TAKE BACKUP:-
# ls linux unix redhat | cpio -ov > /opt/file.cpio
# cd /opt
# ls (to check)
```

```
TO READ BACKUP FILE:-
# cpio -tv < file.cpio
# cd (go to original location & remove files)
# rm -rf linux unix redhat
# ls (to check)
```

```
TO RESTORE:-
# cpio -iv < /opt/files.cpio
# ls (to check)
```

'TO TAKE THE BACKUP FROM DUMP - FOR PARTITION'

```
# mount (to check the partition mount points)
(For Example:- /dev/sda7 mounted on /home)
```

```
TO TAKE THE BACKUP:-
# dump -0uf <destination/backupfile> <partition_no>
# dump -0uf /opt/fullbackup /dev/sda7
# cd /opt
# ls (to check)
```

```
TO READ BACKUP FILE:-
# restore -tf fullbackup
# cd /home (go to the partition mount point)
# rm -rf * (remove all contents)
# ls (to check)
```

```
TO RESTORE:-
# restore -rf /opt/fullbackup
# ls (to check)
```

```
'TO TAKE THE BACKUP FROM SCP PROGRAM - FROM REMOTE PC'  
# ifconfig (to check the ip address)  
# scp -rv <source> <remote_pc_ip:location>  
# scp -rv zoom 192.168.0.10:/media (for example)  
To check the data-Go to the Remote Pc & check in /media  
# rm -rf zoom (remove data from source pc)  
TO RESTORE:-  
# scp -rv <remote_pc_ip:location> <source>  
# scp -rv 192.168.0.10:/media/zoom /root  
# ls (to check)
```

(11)COMPRESSING TOOLS PRACTICAL STEPS

COMPRESSING TOOLS

```
(1)Create one Directory with Data in /root  
(2)Convert the Directory into .tar extension file  
# tar -cvf /opt/zoom.tar zoom (for example)  
# cd /opt  
# ls (to check)
```

TO COMPRESS WITH GZIP:-

```
# ll zoom.tar (check the size)  
# gzip zoom.tar  
# ll zoom.tar.gz (check the size)
```

TO UNCOMPRESS WITH GZIP:-

```
# gunzip zoom.tar.gz  
# ls (to check)
```

TO COMPRESS WITH BZIP2:-

```
# bzip2 zoom.tar  
# ll zoom.tar.bz2 (check the size)
```

TO UNCOMPRESS WITH BZIP2:-

```
# bunzip2 zoom.tar.bz2  
# ls (to check)
```

By:- Sandeep Singh
Mobile No:- +91 9849238833